

Statistics II: Introduction to Inference

Topic 7: Large Sample Theory

March 13, 2025

Question 1: What happens to an estimator if we increase the sample size, i.e. $n \rightarrow \infty$?

Question 2: What is the connection between the sample mean $\bar{X}_n$ and the population mean μ if $n \rightarrow \infty$?

Question 3: What is the sampling distribution of the sample mean $\bar{X}_n$ if $n \rightarrow \infty$?

Example: Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1)$. How do you answer the previous questions!

Convergence in Probability

A sequence of random variables, $X_1, X_2, \dots$, *converges in probability* to a random variable X if, $\forall \epsilon > 0$,

$$\mathbb{P}(|X_n - X| \geq \epsilon) \rightarrow 0 \text{ as } n \rightarrow \infty,$$

or, we can write it as,

$$X_n \xrightarrow{P} X.$$

- If $\mathbb{P}(X = c) = 1$, for any constant c , then we say $X_n \xrightarrow{P} c$.

Sufficient conditions: If $\mathbb{E}(X_n) \rightarrow c$ and $\mathbb{V}(X_n) \rightarrow 0$ as $n \rightarrow \infty$, then $X_n \xrightarrow{P} c$.

Example: Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Ber}(p)$, where $p \in (0, 1)$, then $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} p$.

Question 1: What happens to an estimator if we increase the sample size, i.e. $n \rightarrow \infty$?

Answer: A nice large sample property of estimators is **Consistency**.

Definition: Let $T_1 = T(X_1)$, $T_2 = T(X_1, X_2)$, $\dots, T_n = T(X_1, \dots, X_n)$ be a sequence of estimators of $\psi(\theta)$, then the sequence of estimators $\{T_n\}$ is said to be consistent if

$$T_n \xrightarrow{P} \psi(\theta).$$

Example: Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Uniform}(0, \theta)$. Let $T_1 = X_1, T_2 = \max\{X_1, X_2\}, \dots, T_n = \max\{X_1, \dots, X_n\} = X_{(n)}$ be a sequence of estimators of θ . Then, $X_{(n)}$ is a consistent estimator of θ .

Example: Suppose $X_1, \dots, X_n$ are independently and identically distributed random samples from $\text{Uniform}(0, \theta)$ distribution, where $\theta > 0$. Show that the following three estimators

- (i) $T_1 = X_{(n)}$,
- (ii) $T_2 = \frac{n+1}{n} X_{(n)}$, &
- (iii) $T_3 = 2\bar{X}_n$

are consistent estimators of $\psi(\theta) = \theta$.

Continued...

```
1  set.seed(2025)
2  n = seq(10, 2500, by = 10)
3  theta = 2  # true parameter
4  x = runif(max(n), 0, theta)
5  T1 = T2 = T3 = c(0)
6  for (i in 1:length(n)) {
7    T1[i] = max(x[1:n[i]])
8    T2[i] = ((n[i]+1)/n[i]) * T1[i]
9    T3[i] = 2*mean(x[1:n[i]]) }
10 est = cbind(T1, T2, T3)
11 # Consistency checking
12 matplot(n, est, type="l", lty=c(2,3,4), ylab="Estimators",
13         xlab="Sample size(n)", col=c("blue","green","red"))
14 abline(h = theta)
15 legend("topright", legend=c("Truth", "T1", "T2", "T3"),
16        col=c("black", "blue", "green", "red"), lty=c(1,2,3,4))
```

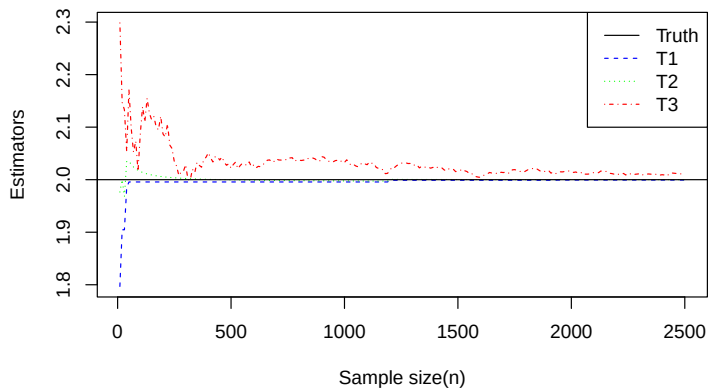


Figure 1: Checking the consistency of T_1 , T_2 and T_3 .

Markov's and Chebyshev's Inequality

Markov's Inequality: Let X be a non-negative random variable whose $\mathbb{E}(X)$ exists. Then, for every real number $t > 0$,

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}(X)}{t}.$$

Chebyshev's Inequality: Let X be a random variable for which $\mathbb{V}(X)$ exists. Then, for every real number $t > 0$,

$$\mathbb{P}(|X - \mathbb{E}(X)| \geq t) \leq \frac{\mathbb{V}(X)}{t^2}.$$

Weak Law of Large Numbers (WLLN)

Question 2: What is the connection between the sample mean $\bar{X}_n$ and the population mean μ if $n \rightarrow \infty$?

Answer: **WLLN**.

Definition of WLLN: Let $X_1, X_2, \dots$ be *iid* random variables with $\mathbb{E}(X_i) = \mu$ and $\mathbb{V}(X_i) = \sigma^2 (< \infty)$. Then,

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mu$$

or, $\bar{X}_n$ is a consistent estimator of μ .

WLLN for independent but non-identical case

Let $X_1, X_2, \dots$ be independent random variables with $\mathbb{E}(X_i) = \mu_i$ and $\mathbb{V}(X_i) = \sigma_i^2$ such that $\max_i \sigma_i^2 < M (< \infty)$.

Then,

$$\bar{X}_n - \bar{\mu}_n \xrightarrow{P} 0$$

where $\bar{\mu}_n = \frac{1}{n} \sum_{i=1}^n \mu_i$.

Convergence in Distribution

As the name suggests, it is intuitively convergence of distribution functions of random variables.

Definition: A sequence of random variables, $X_1, X_2, \dots$, *converges in distribution* to a random variable X , if

$$\lim_{n \rightarrow \infty} F_n(t) = F_X(t)$$

at all points t where $F_X(t)$ is continuous. Here F_n is the CDF of X_n and F_X is the CDF of X .

Generally, we write it as,

$$X_n \xrightarrow{d} X.$$

Example: Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Uniform}(0, 1)$. Then,
 $n(1 - X_{(n)}) \xrightarrow{d} \text{Exponential}(1)$.

Central Limit Theorem (CLT)

Question 3: What is the sampling distribution of the sample mean $\bar{X}_n$ if $n \rightarrow \infty$?

Answer: **CLT**.

Definition of CLT: Let $X_1, X_2, \dots$ be *iid* random variables with $\mathbb{E}(X_i) = \mu$ and $\mathbb{V}(X_i) = \sigma^2 < \infty$. Then,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1) \text{ as } n \rightarrow \infty,$$

or, $\forall x \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq x \right) = \Phi(x),$$

where $\Phi(\cdot)$ is the CDF of the standard normal distribution.

Example: (Continuous) Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Uniform}(0, \theta)$, where $\theta > 0$. Check WLLN and CLT through simulation.

Solution: $\mathbb{E}(X_1) = \theta/2$ and $\mathbb{V}(X_1) = \theta^2/12$.

- By WLLN, $\bar{X}_n \xrightarrow{P} \theta/2$
- By CLT, $\frac{\sqrt{n}(\bar{X}_n - \theta/2)}{\theta/\sqrt{12}} \xrightarrow{d} N(0, 1)$

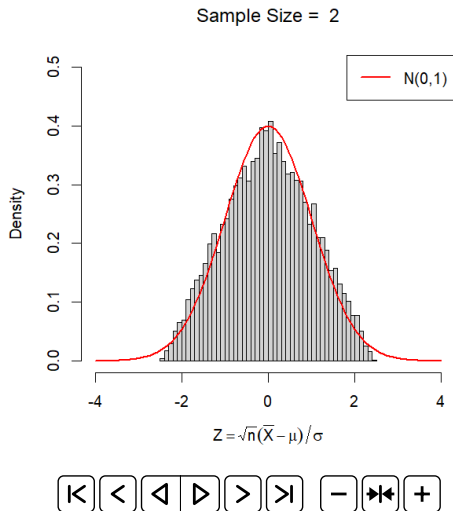
Example: (Discrete) Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poisson}(\lambda)$, where $\lambda > 0$. Check WLLN and CLT through simulation.

Solution: Do the exercise as before.

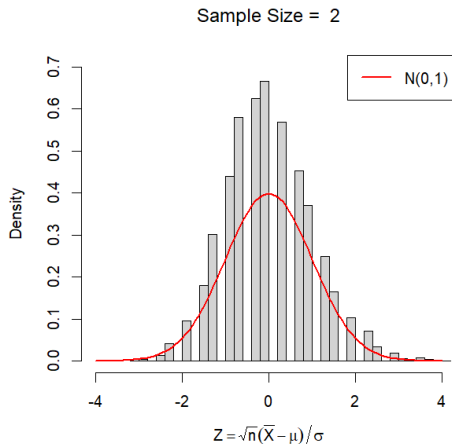
R code for CLT

```
1 repeat = 10000; n = 1000; xbar = c(0)
2 theta = 2 # true parameter
3
4 z.values = replicate(replication, {
5   sample = runif(n, 0, theta) # Draw n samples
6   xbar = mean(sample) # Compute sample mean
7   (sqrt(n) * (xbar - (theta/2))) / (theta/sqrt(12))
8 })
9
10 hist(z.values, probability = TRUE, breaks = 40,
11      main = bquote("Sample Size = " ~ .(n)),
12      xlab = expression(Z == sqrt(n) * (bar(X) - mu) / sigma))
13
14 curve(dnorm(x, mean = 0, sd = 1),
15      col = "red", lwd = 2, add = TRUE)
```

Histogram Plot for Uniform(0, θ)



Histogram Plot for Poisson(λ)



Exercises

1. Suppose that an *iid* random sample $X_1, \dots, X_n$ of size $n = 12$ is taken from $U(0, 1)$. Find $\mathbb{P}(|\bar{X}_n - \frac{1}{2}| \leq 0.1)$ using CLT, where $\bar{X}_n$ is the sample mean.
2. Let $X_1, \dots, X_n$ be an *iid* random sample from the $\text{Poisson}(\theta)$ distribution. Also, let $\bar{X}_n$ be the sample average. Find the value of c for which $\mathbb{P}(|\bar{X}_n - \theta| < c) \approx 0.5508$ using CLT, where $n = 55$ and $\theta = 7$.
3. Suppose that a random sample of size n is to be taken from a distribution for which the mean is μ and the standard deviation is 3. Use the central limit theorem to determine approximately the smallest value of n for which the following relation will be satisfied

$$\mathbb{P}(|\bar{X}_n - \mu| < 0.3) \geq 0.95.$$

4. Let T_n be a consistent sequence of estimators of θ , and $\{a_n\}$ and $\{b_n\}$ be two sequences of constants such that $\lim_{n \rightarrow \infty} a_n = 1$ and $\lim_{n \rightarrow \infty} b_n = 0$. Show that $a_n T_n + b_n$ is also a consistent sequence of estimators of θ .
5. Let $X_1, \dots, X_n$ be a sequence of *iid* random variables, and $c_1 < c_2$ be two constants. Define $p = \mathbb{P}(c_1 \leq X_1 < c_2)$. Now, we only know whether an observation X_i belongs to $[c_1, c_2)$ or not. How do you approximate the value of p for $c_1 = 1$ and $c_2 = 2$?
6. Suppose that a fair coin is tossed 900 times. Find an approximate probability of obtaining more than 495 heads. How do you calculate the same for a biased coin (i.e. $\mathbb{P}(\text{Head}) = p$ (unknown))?